School of Computer Science and Engineering
Dept. of Computer Science and Engineering
COURSE PLAN
Academic Year 2025-26 EVEN SEMESTER

School/Department of Students	**PSCS**
Name of the Program(s) of Students	B.Tech. Computer Science and Technology Big Data (CBD)
PRC Approval Ref. No.	PU/AC-21.5/SoCSE2/CBD/2023-2027
Semester/Year	VI/III
Course Code & Name	CSE2263 Analysis of Algorithms Lab
Credit Structure (L-T-P-C)	0 – 0 – 2 – 1
Contact Hours	2 Sessions per week -30 Sessions
Course In-Charge (IC)	Dr. S. Aarif Ahamed, Dr. Praveena and Dr. Madhusudhan M V
Course Instructor(s)	1. Dr. S. Aarif Ahamed 2. Mr. Anandan B 3. Mr. Asad Mohammed Khan 4. Ms. Babitha 5. Mr. Dinesh C 6. Ms. Chandana 7. Mr. Girish Kumar 8. Dr. Jagadevi Bakka 9. Mr. Jaikumar 10. Dr. Manjula H M 11. Ms. Meena Kumari K S 12. Mr. Mohammed Shakir 13. Ms. Padma T G 14. Dr. Pallavi R 15. Ms. Sandhya L 16. Mr. Shankar J 17. Dr. Shanthi S 18. Mr. Sheikh Wasim Masood 19. Ms. Shet Reshma 20. Dr. Sridevi S 21. Ms. Suma N G 22. Dr. Yamanappa 23. Dr. Pamela Vinitha 24. Ms. Anu Joseph 25. Dr. Praveena k N 26. Dr. Jayachandran A 27. Dr. Ruhin Kouser 28. Mr. Katta Sriram 29. Ms. Priyanka Gour 30. Dr. Sandeep Albert Mathias 31. Mr. Sakthi S

	32. Dr. Madhusudhan M V 33. Ms. Dumpala Shwetha 34. Ms. Radha K 35. Mrs. Sharanya S Salian 36. Ms. Subhashini R 37. Ms. Shana Aneevan 38. Ms. Swetha K H
Course URL	https://presidencyuniversity.linways.com

1. COURSE PRE-REQUISITES:
NIL

2. COURSE DESCRIPTION:
This course introduces techniques for the design and analysis of efficient algorithms and methods of applications. This course discusses the classic approaches for algorithm design such as Divide and Conquer, Dynamic Programming, Greedy method. This course also describes other basic strategies searching solution space. The core concepts of analyzing algorithms and classifying them into various complexity classes is covered in the end.

3. COURSE OBJECTIVES:
The objective of the course is to familiarize the learners with the concepts of Analysis of Algorithms and attain Skill Development through Experiential Learning Methodologies.

4. COURSE OUTCOMES:

TABLE 1: COURSE OUTCOMES

CO Number	**Statement of CO**	**Blooms Cognitive Level**
	On successful completion of the course the students shall be able to	
CO1	Compute efficiency of a given algorithm	Apply
CO2	Apply divide and conquer technique for searching and sorting Problems	Apply
CO3	Apply the Dynamic Programming technique for a given problem	Apply
CO4	Apply greedy technique for solving a Problem	Apply
CO5	Demonstrate Back tracking technique and limitations of Algorithms	Apply

5. MAPPING OF COURSE OUTCOMES WITH PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMES:

5.1 *PROGRAM OUTCOMES*:

(A new set of POs, if any, should be used for the courses offered to the students admitted in the 2025 batch.)

On successful completion of the Program, the students will be able to:

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

TABLE 2a: CO-PO Mapping												
CO. No	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	H	M	-	M	H	-	-	-	-	-	-	H
CO2	H	M	-	M	H	-	-	-	-	-	-	H
CO3	H	M	-	M	H	-	-	-	-	-	-	H
CO4	H	M	-	M	H	-	-	-	-	-	-	H
CO5	H	M	-	M	H	-	-	-	-	-	-	H

5.2 PROGRAM SPECIFIC OUTCOMES:

On successful completion of the Program, the students will be able to:

(New Set of PSOs, if any, needs to be used)

PSO1	Problem Analysis: Identify, formulate, research literature, and analyse complex engineering problems related to Software Engineering principles & practice, Programming, Big Data computing & analytics Substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PSO2	Design/development of Solutions: Design solutions for complex engineering problems related to Software Engineering principles & practice, Programming, Big Data Computing & analytics and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PSO3	Modern Tools Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities related to Software Engineering principles & practice, Programming, Big Data Computing & analytics with an understanding of the limitations.

TABLE 2b: CO-PSO Mapping			
CO Number	**PSO1**	**PSO2**	**PSO3**
CO1	M	-	H
CO2	M	-	H
CO3	M	-	H
CO4	M	-	H
CO5	M	-	H

6. COURSE CONTENT:

Module Number	Module Name	Number of Sessions

Lab Sheet 1	Measuring running time of an algorithm	1
Lab Sheet 2	Compare running time of algorithms	1
Lab Sheet 3	Implement sorting algorithms: Bubble Sort and Selection Sort	1
Lab Sheet 4	Compare searching algorithms: Linear Search and Binary Search	1
Lab Sheet 5	Compare sorting algorithms: Insertion Sort and Merge Sort	1
Lab Sheet 6	Quick Sort and analysis of pivot selection	1
Lab Sheet 7	Implementing Factorial and Coin Change problem using Dynamic Programming	1
Lab Sheet 8	0/1 Knapsack Problem using Dynamic Programming	1
Lab Sheet 9	Floyd-Warshall's Algorithm	1
Lab Sheet 10	Fractional Knapsack Problem using Greedy technique	1
Lab Sheet 11	Minimum Spanning Tree using Prim's Algorithm	1
Lab Sheet 12	Minimum Spanning Tree using Kruskal's Algorithm	1
Lab Sheet 13	Knapsack Problem using Branch and Bound technique	1

Lab Sheet 14	N-Queens Problem using Backtracking	1
Lab Sheet 15	Case Study: Knapsack problem using multiple techniques	1

REFERENCE MATERIALS:

Text Books:

T1. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", 3rd edition, Pearson Education, 2018.

T2. Thomas H.Cormen, Charles E.Leiserson, Ronald L. Rivest and Clifford Stein, "Introduction to Algorithms", 4th edition, MIT Press, 2022.

Reference Books:

R1. J. Kleinberg and E. Tardos, "Algorithm Design", Addison-Wesley, 2005.

R2. Tim Roughgarden, "Algorithms Illuminated" (books 1 through 3), "Operating Systems Design and Implementation", Sound like yourself Publishing, 2017-2019.

R3. AV Aho, J Hopcroft, JD Ullman, "The Design and Analysis of Algorithms", Addison-Wesley, 1974.

R4. Donald E. Knuth, "The Art of Computer Programming", Volumes 1and 3 Pearson.

Online Resources

W1. NPTEL: https://onlinecourses.nptel.ac.in/noc26_cs67/preview

W2. Coursera: Analysis of Algorithms by Princeton University

W3. Algorithms Specialization in Coursera by Stanford University(Group of 4 courses).

W4. Algorithms Coding Contest Links maintained by Prof Gerth Stølting Brodal of Aarhus University

7. DETAILED SCHEDULE OF INSTRUCTION

TABLE 3: LESSON PLAN				
Session Number	**Topic**	**Sub-Topic**	**CO Number**	**Reference**
LabSheet-1	Program Integration & Course Integration	Course overview, algorithm paradigms, lab tools & evaluation; execution time measurement for varying input sizes	CO1	T1 (Ch.1, Pg: 1–10)
Module 1				
LabSheet-2	Compare Running Time of Algorithms	Comparative evaluation of algorithm performance for large input sizes	CO1	T1 (Ch.1, Pg: 11–18)
LabSheet-3	Sorting Algorithms	Implementation of Bubble Sort and Selection Sort	CO1	T1 (Ch.1, Pg: 19–28)
Module 2				

LabSheet-4	Searching Algorithms	Linear Search and Binary Search – implementation and comparison	CO2	T1 (Ch.2, Pg: 43–44)
LabSheet-5	Sorting Algorithms	Insertion Sort and Merge Sort – performance comparison	CO2	T1 (Ch.2, Pg: 45–60)
LabSheet-6	Quick Sort	Pivot selection strategies and impact on running time	CO2	T1 (Ch.2, Pg: 61–72)
Module 3				
LabSheet-7	Dynamic Programming	Factorial & Coin Change Problem using Dynamic Programming	CO3	T1 (Ch.3, Pg: 85–92)
LabSheet-8	Dynamic Programming	0/1 Knapsack Problem using Dynamic Programming	CO3	T1 (Ch.3, Pg: 93–102)
LabSheet-9	Dynamic Programming	Floyd-Warshall's Algorithm – all-pairs shortest path	CO3	T1 (Ch.3, Pg: 103–112)
Module 4				
LabSheet-10	Greedy Technique	Fractional Knapsack Problem	CO4	T1 (Ch.4, Pg: 125–132)
LabSheet-11	Greedy Technique	Minimum Spanning Tree using Prim's Algorithm	CO4	T1 (Ch.4, Pg: 133–140)
LabSheet-12	Greedy Technique	Minimum Spanning Tree using Kruskal's Algorithm	CO4	T1 (Ch.4, Pg: 141–148)
Module 5				
LabSheet-13	Branch and Bound	Knapsack Problem using Branch and Bound technique	CO5	T1 (Ch.5, Pg: 160–168)
LabSheet-14	Backtracking	N-Queens Problem using Backtracking	CO5	T1 (Ch.5, Pg: 169–176)
LabSheet-15	Case Study	Knapsack problem solved using Greedy, DP & Branch and Bound	CO5	T1 (Ch.5, Pg: 177–182)

The main pedagogical methods in the course are as follows:

- Lecture mode.
- PowerPoint Presentation.
- Video-based learning.
- Problem-based learning method.
- Simulation Practical system case study/Model Design.

TABLE 4: SPECIAL DELIVERY METHOD			
S. No	**Session Number**	**Subtopic (as per lesson plan)**	**Pedagogical Method**

1.	5	Insertion Sort and Merge Sort – performance comparison	Problem-based learning method
2.	9	Floyd-Warshall's Algorithm	Video-based learning + Simulation
3.	14	N-Queens Problem using Backtracking	Problem-based learning + Model Design
4.	15	Case Study: Knapsack problem using multiple techniques	Simulation Practical system case study / Model Design

8. ASSESSMENT SCHEDULE

TABLE 5: ASSESSMENT SCHEDULE

Sl. No	Assessment Type	Coverage	CO Number(s)	Duration in Minutes	Marks	Weightage
1.	CA 1	Lab Sheet 1 - 3	CO1	50	40	20%
2.	Midterm Exam	Lab Sheet 1 - 6	CO1 & CO2	100	50	25%
3.	CA 2	Lab Sheet 7 - 9	CO3	50	40	20%
4.	CA 3	Lab Sheet 10 - 12	CO4	50	20	10%
5.	End Term Examination	Lab Sheet 1 - 15	CO1, CO2, CO3, CO4 & CO5	100	50	25%

9. COURSE CLEARANCE CRITERIA:

This is in accordance with the Academic Regulations of the University and the Program Regulations and Curriculum of the respective program.

10. SAMPLE QUESTIONS:

TABLE 6: SAMPLE QUESTIONS

Sl. No	Question	Marks	CO Number	Blooms Cognitive Level
1	Write a program to measure the running time of an algorithm for different input sizes and analyze the results.	10	CO1	Apply
2	Write a program to implement Bubble Sort and Selection Sort and compare their performance.	10	CO2	Apply
3	Implement Factorial using Dynamic Programming and compare it with the recursive approach.	10	CO3	Apply
4	Develop a program to solve the Fractional Knapsack problem using the Greedy method.	10	CO4	Apply
5	Implement the N-Queens problem using the Backtracking technique and explain the solution process.	10	CO5	Apply

11. MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs):

TABLE 7: SDG MAPPING			
S. No	**Topic**	**SDG Number**	**Justification**
1	Measuring Running Time of Algorithms	SDG 4 – Quality Education	Enhances computational thinking and analytical skills through experiential learning, improving quality of technical education.
2	Searching and Sorting Algorithms	SDG 9 – Industry, Innovation and Infrastructure	Efficient searching and sorting form the foundation for scalable software systems and digital infrastructure.
3	Dynamic Programming Techniques	SDG 9 – Industry, Innovation and Infrastructure	Optimization techniques support innovative solutions in engineering and technology applications.
4	Greedy Algorithms (Knapsack, MST)	SDG 8 – Decent Work and Economic Growth	Optimization algorithms help improve resource utilization and operational efficiency in industries.
5	Branch and Bound & Backtracking Techniques	SDG 4 – Quality Education	Promotes advanced problem-solving and logical reasoning skills essential for higher education and research.

12. CRITERIA FOR COURSE OUTCOME ATTAINMENT CALCULATION:

TABLE 8: Threshold and Target Set for Course Outcomes				
Sl. No	**C.O. No.**	**Course Outcomes**	**Threshold in %**	**Target in %**
1.	CO1	Compute efficiency of a given algorithm	60	65
2.	CO2	Apply divide and conquer technique for searching and sorting Problems	60	65
3.	CO3	Apply the Dynamic Programming technique for a given problem	60	65
4.	CO4	Apply greedy technique for solving a Problem	60	65
5.	CO5	Demonstrate Back tracking technique and limitations of Algorithms	60	65

13. SUMMARY:

TABLE 9: SUMMARY OF COURSE SCHEDULE				
Sl. No.	**Activity**	**Start date**	**End date**	**Total number of Sessions**
1.	Program Integration & Over View of the course	07.01.2026	07.01.2026	1
2.	Module 1: Introduction (Lab Sheets 1–3)	08.01.2026	08.02.2026	3
3.	Test – 1 (Lab Sheets 1–3)	09.02.2026	14.02.2026	1
4.	Module 2: Divide and Conquer (Lab Sheets 4–6)	15.02.2026	09.03.2026	3
5.	Midterm Examination (Lab Sheets 1–6)	10.03.2026	14.03.2026	1
6.	Module 3: Dynamic Programming (Lab Sheets 7–9)	15.03.2026	26.03.2026	3
7.	Test – 2 (Lab Sheets 7–9)	27.03.2026	01.04.2026	1

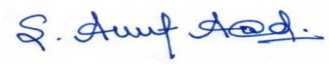
8.	Module 4: Greedy Technique (Lab Sheets 10–12)	02.04.2026	16.04.2026	3
9.	Test – 3 (Lab Sheets 10–12)	17.04.2026	28.04.2026	1
10.	Module 5: Complexity Classes (Lab Sheets 13–15)	29.05.2026	06.05.2026	3
11.	End Term Examination (Lab Sheets 1–15)	06.05.2026 onwards		1

CONTACT TIMINGS IN THE CHAMBER FOR DISCUSSION

Students can meet the respective course instructor during the Chamber Consultation Hour to clarify doubts related to the course.

SPECIFIC GUIDELINES TO STUDENTS, IF ANY:

- Attend all classes regularly.
- Bring a scientific calculator to every class.
- Refer to online study materials and watch the suggested videos available on the NPTEL website.



Name and Signature of the course In-Charge

Dr. S. Aarif Ahamed

APPROVAL:

This course has been duly verified and approved by the Departmental Academic Committee (DAC).

Name and Signature of the Chairperson - DAC